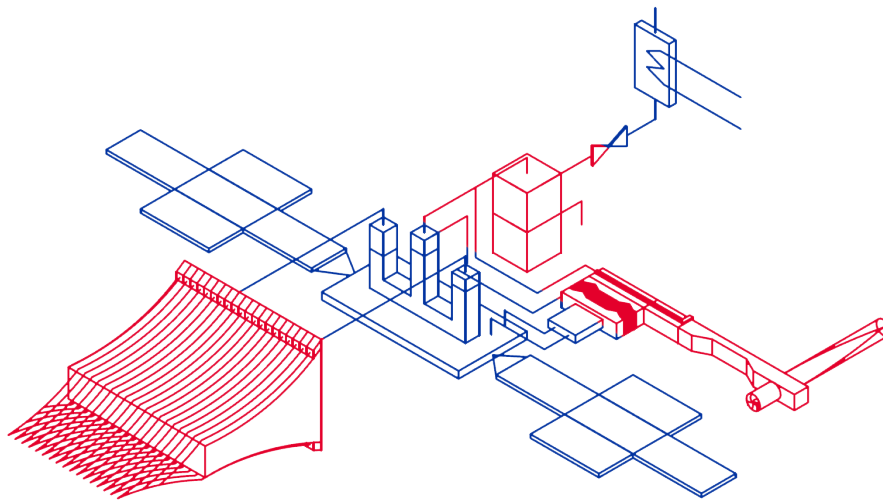


ICER Microwave Plasma Thruster for Skyraker™ Technical Demo

Complete Theoretical, Thermodynamic and Practical Guide

by Giovanni Nicola D'Aloisio



Demonstrator Concept

Microwave source: 1 kW magnetron salvaged from domestic microwave oven (2.45 GHz)

Waveguide: Handmade from aluminium flat bar / sheet, 600×90×50 mm, flattened to 25 mm at ionisation cavity

Plasma tube: Borosilicate/quartz tube, $\varnothing_i = 24$ mm, L=600 mm

Air supply: Compressed air (compressor) fed as vortex into plasma tube

Thrust measurement: Steel ball force gauge (after Ye et al.) + optional torsion balance

Estimated cost: ≈€50–€150 (magnetron: salvage; waveguide: DIY)

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Part of the Space Framework™

Based on D. Ye, J. Li, J. Tang, *Jet propulsion by microwave air plasma in the atmosphere*, AIP Advances **10**, 055002 (2020)



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1 Overview and Motivation

The ICER (Ion Cruise and Exospheric-breathing Rocket) subsystem provides the Skyraker high- I_{sp} electric propulsion for interplanetary cruise and atmospheric-breathing orbit maintenance.

Phase 5 builds a **microwave air plasma thruster** operating at atmospheric pressure, following the design published by Ye et al. (2020) at Wuhan University. This approach has two advantages:

1. **No vacuum chamber required.** The thruster operates in air, making it the only plasma propulsion technology demonstrable in a standard university laboratory.
2. **Zero-cost ionisation source.** The magnetron is salvaged from a domestic microwave oven; the waveguide is fabricated by hand from aluminium flat bar.

The Dan Ye thruster uses atmospheric air as propellant, ionised by microwave energy, which is precisely the ABEP concept of Skyraker. The Phase 5 demonstrator validates the ionisation-and-jet principle at bench level. The transition to space-grade ABEP (vacuum, low-pressure tenuous atmosphere) is a **TRL 5** milestone.

2 Operating Principle

2.1 Microwave Air Plasma Jet

A 2.45 GHz microwave field is concentrated in a flattened waveguide section. A quartz tube passes vertically through the cavity. Compressed air enters the tube from the side as a **vortex**, which:

- stabilises the plasma axially (centrifugal confinement);
- protects the quartz wall from the hot plasma core ($T > 1000^\circ\text{C}$);
- provides the propellant mass flow $\dot{m}$.

The microwave energy ionises the air, creating a high-temperature, high-pressure plasma torch. The plasma expands through the open end of the tube, generating thrust.

2.2 Thrust and Jet Pressure

Following Ye et al. (2020), the net propulsion force due to the plasma jet (excluding the contribution from the injected air at zero microwave power):

$$F_{\text{net}} = F - F_0 = (M - M_0)g \quad (1)$$

where M is the critical threshold weight of the steel ball (the minimum total weight that keeps the ball stationary on top of the tube), M_0 is the same quantity at zero microwave power.

The net jet pressure over the quartz tube cross-section:

$$P_{\text{net}} = \frac{F - F_0}{\pi R^2} \quad (2)$$

where $R = 12\text{ mm}$ (inner radius of quartz tube, $\varnothing_i = 24\text{ mm}$).

2.3 Propulsive Efficiency and I_{sp}

Unlike a conventional ion thruster, the microwave plasma jet does not electrostatically accelerate ions to a single well-defined exit velocity. Instead, it heats the gas thermally, producing a jet with a Maxwellian-like velocity distribution. The effective exhaust velocity:

$$v_{\text{eff}} = \frac{F_{\text{net}}}{\dot{m}_{\text{air}}}, \quad I_{sp, \text{meas}} = \frac{v_{\text{eff}}}{g_0} = \frac{F_{\text{net}}}{\dot{m}_{\text{air}} g_0}, \quad \eta_{\text{prop}} = \frac{F_{\text{net}}^2}{2 \dot{m}_{\text{air}} P_\mu} \quad (3)$$

where P_μ is the microwave power absorbed by the plasma.

2.4 Scaling Law from Ye et al.

The experimental data of Ye et al. exhibit a linear relationship between propulsion force and microwave power at constant air flow:

$$F_{\text{net}}(P_{\mu}, \dot{V}) = m_{\text{fit}}(\dot{V}) \cdot P_{\mu} + c_{\text{fit}}(\dot{V}) \quad (4)$$

At 400 W, 1.45 m³/h: $F_{\text{net}} \approx 11$ N; lifting force ≈ 28 N/kW; $P_{\text{net}} \approx 2.4 \times 10^4$ N/m².

P_{μ} [W]	$\dot{V}$ [m ³ /h]	F [N]	$F_{\text{net}} = F - F_0$ (air only) [N]	F/P [N/kW]
400	1.45	11.0	7.0	28
600	1.15	12.0	7.5	20
800	1.00	14.0	9.5	17

Table 1: Representative results from Ye et al. (2020) used to calibrate the Phase 5 demonstrator.

3 Waveguide Design and Fabrication

3.1 Rectangular Waveguide Theory

For the dominant TE₁₀ mode in a rectangular waveguide of width a and height b :

$$\lambda_c = 2a \quad (\text{cutoff wavelength}) \quad (5)$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - (\lambda_0/\lambda_c)^2}} \quad (\text{guide wavelength}) \quad (6)$$

$$f_c = \frac{c}{2a} \quad (\text{cutoff frequency}) \quad (7)$$

At 2.45 GHz ($\lambda_0 = 122.4$ mm), the standard WR-340 waveguide has $a = 86.36$ mm, $b = 43.18$ mm. For a handmade waveguide, the critical constraint is $a > \lambda_0/2 = 61.2$ mm.

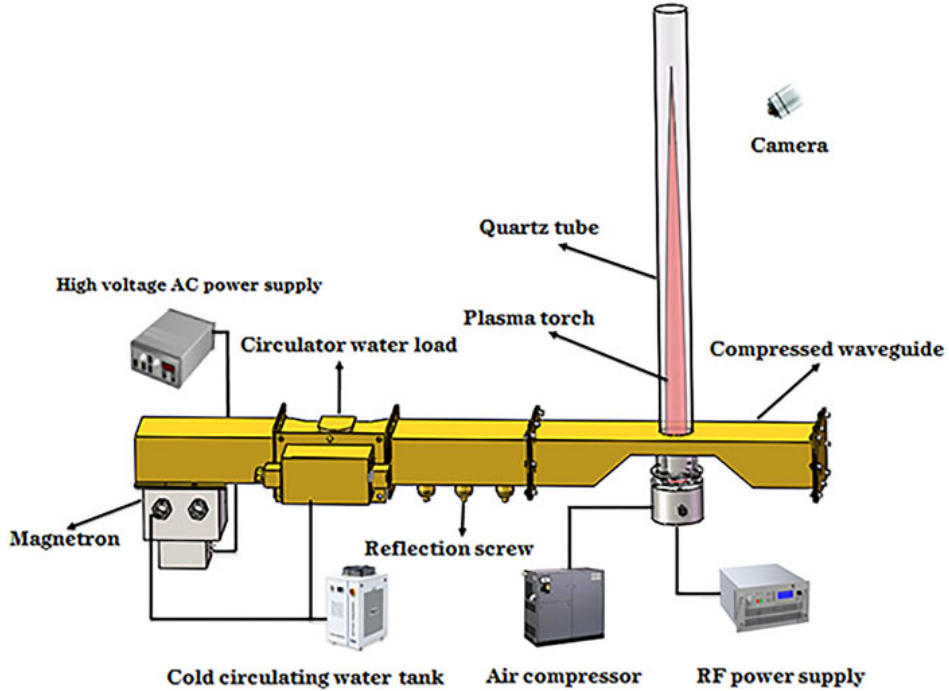


Figure 1: Microwave plasma thruster schematic from [6].

3.2 Handmade Waveguide – Fabrication

Following Ye et al. (2020), the waveguide is fabricated by hand from aluminium flat bar (grade 6061 or 5083, thickness ≥ 2 mm). The quartz tube is positioned at $\lambda_g/4$ from the short-circuit end.

Section	Dimension	Value	Unit
Main waveguide (full height)	$a \times b \times L$	90×50×600	mm
Flattened cavity section	$a \times b_f \times L_f$	90×25×100	mm
Quartz tube position	from short end	$\lambda_g/4$	mm
Quartz tube aperture (circular)	diameter	27	mm
Three-stub tuner position	–	$\lambda_g/4$ from cavity	mm
Short-circuit plate (end)	–	full cross-section	–

Table 2: ICER waveguide dimensions (after Ye et al. 2020).

3.3 Step-by-Step Waveguide Construction

WG1. Cut aluminium flat bar. Two pieces $90 \times 600 \times 2$ mm (top and bottom); two pieces $50 \times 600 \times 2$ mm (sides). These form the main rectangular body.

WG2. Bend and join. Fold and rivet or screw the four sides together, ensuring good electrical contact at all seams (use conductive tape or solder/braze joints if possible). A gap at a seam reflects microwave power and reduces efficiency; total gap < 1 mm acceptable.

WG3. Fabricate the flattened cavity section. At position $L = \lambda_g/4$ from the short end, press-form the top and bottom walls to reduce height from 50 mm to 25 mm over a 100 mm length. Use a vice and steel form block.

WG4. Drill the quartz tube hole. Drill a 28 mm circular hole through the top and bottom of the flattened section, centred at $\lambda_g/4$ from the short-circuit end. File the edge smooth: no burrs that could arc.

WG5. Calculate $\lambda_g/4$. Since this, position quartz tube at 44 mm from the short-circuit end plate:

$$\lambda_g = \frac{122.4 \text{ mm}}{\sqrt{1 - (122.4/180)^2}} \approx 176 \text{ mm} \quad \Rightarrow \quad \lambda_g/4 \approx 44 \text{ mm} \quad (8)$$

WG6. Make the short-circuit plate. Cut a 90×50 mm aluminium plate; screw to the end of the waveguide. This reflector sets up a standing wave that maximises the electric field at the quartz tube position.

WG7. Three-stub tuner. Drill three $\varnothing 8$ mm holes in the top wall, spaced $\lambda_g/4$ apart. Insert M8 stainless steel bolts as stubs; adjust depth to minimise reflected power (use SWR meter or minimise heating of circulator load).

WG8. Magnetron coupling. Remove the magnetron from the donor microwave oven. The magnetron antenna (probe) extends into the waveguide via a coupling hole in the broad wall. Retain the original magnetron door interlock: modify it so the system operates only when a specific switch is closed (safety interlock).

3.4 Guide Wavelength Calculation

For $a = 90$ mm, $f = 2.45$ GHz:

$$f_c = \frac{c}{2a} = \frac{3 \times 10^8}{0.180} = 1.667 \text{ GHz} \quad (9)$$

$$\lambda_g = \frac{c/f}{\sqrt{1 - (f_c/f)^2}} = \frac{122.4}{\sqrt{1 - (1.667/2.45)^2}} = \frac{122.4}{\sqrt{0.537}} \approx 167 \text{ mm} \quad (10)$$

$$\lambda_g/4 \approx 42 \text{ mm} \quad (11)$$

4 Magnetron – Domestic Microwave Salvage

4.1 Which Microwave Oven to Use

Standard domestic microwave ovens use a **2M2xx series magnetron** (Panasonic, LG, Samsung – all interchangeable for this application) operating at 2.45 GHz with output power 700–1200 W.

Any microwave oven ≥ 800 W with a replaceable magnetron (avoid sealed units) is suitable. Recommended: 1000 W unit from a second-hand market or university surplus store.

4.2 Magnetron Extraction

- M1. Discharge capacitor before handling.** The HV capacitor (≈ 2100 V) retains charge for hours after power-off. Short the capacitor terminals with an insulated screwdriver before touching any component. **This step is non-negotiable.**
- M2. Remove magnetron.** Unscrew the magnetron from the waveguide flange (4 screws, M3 or M4). Retain the cooling fan; it will be reused.
- M3. Retain the transformer.** The microwave oven transformer (MOT) provides the magnetron HV supply (≈ 2100 V at 300 mA). It is salvaged and reconnected to the magnetron in the new assembly.
- M4. Retain the door interlock switches.** Wire the interlock switches in series with the MOT primary. Without this, the system can be energised with the plasma torch accessible to the operator – a lethal hazard.
- M5. Construct the circulator (water load).** A circulator is needed to absorb reflected microwave power that could damage the magnetron. A simple water load can be made from a sealed plastic container of water (≈ 1 L) placed in the reflected-wave path. A ferrite circulator from a scrapped microwave communications unit is preferable.

Budget note. A complete magnetron assembly (magnetron + transformer + HV capacitor + diode) from a salvaged microwave oven costs €0–€20. A new 2.45 GHz magnetron module costs €30–€60 from electronics suppliers. For this phase, the salvage route is preferred: the oven provides magnetron, HV supply, cooling fan, and a donor waveguide section all in one unit.

5 Plasma Tube and Air Circuit

5.1 Quartz vs. Borosilicate Tube

Material	T_{cont} [°C]	T_{max} [°C]	Cost ($\varnothing_i = 24$, L=600 mm)	Notes
Fused silica (quartz)	1100	1700	≈€40	Best; low dielectric loss
Borosilicate (DURAN)	500	800	≈€15	Acceptable; check for hot spots
Alumina (Al_2O_3)	1600	1750	≈€60	Opaque; preferred for long runs

Table 3: Plasma tube material comparison.

Recommendation: Use borosilicate glass for initial tests (cheaper, easier to source). Upgrade to quartz if localised hot spots cause cracking. The vortex air flow protects the wall, so borosilicate survives at moderate power levels (400–600 W).

5.2 Air Circuit

1. **Air compressor:** any lab compressor delivering ≥ 30 L/min at 1–3 bar gauge pressure.
2. **Flow meter:** rotameter 0–2 m³/h (or 0–33 L/min).
3. **Needle valve:** fine flow control.
4. **Vortex inlet fitting:** a tangential elbow fitting on the side of the quartz tube, pointing circumferentially to induce vortex rotation. Fabricate from a bent 6 mm copper tube soldered to a bored SS316 collar.

6 Thrust Measurement System

6.1 Method 1 – Steel Ball Gauge (after Ye et al.)

- T1-1.** Machine or find a hollow steel ball, mass M_{ball} , outer diameter $D_{\text{ball}} > \varnothing_{\text{tube}}$ (so it rests on the tube opening).
- T1-2.** Drill a small hole ($\varnothing \approx 5$ mm) in the top of the ball for inserting calibrated steel beads.
- T1-3. No microwave ($P_{\mu} = 0$):** Set air flow to target value; add beads until ball is stationary. Total mass = M_0 ; $F_0 = M_0 g$.
- T1-4. With microwave ($P_{\mu} > 0$):** Turn on magnetron; add beads until ball is stationary again. Total mass = M ; $F = M g$.
- T1-5.** Net plasma thrust: $F_{\text{net}} = (M - M_0)g$.

Why the ball must vibrate. The threshold condition – minimum weight to keep the ball *stationary* – is the maximum static friction of the jet. When the ball is at threshold, it begins to vibrate (rattle) from turbulent jet fluctuations. This is the observable criterion. The measurement protocol is: add beads one by one until vibration ceases, then record total mass.

6.2 Method 2 – Load Cell (optional upgrade)

Replace the steel ball gauge with a rigid aluminium piston (same diameter as tube) resting on the plasma exit, connected to the 100 g load cell via a thin wire. This gives continuous force recording at 10 Hz via the Arduino DAQ system from Phase 3A (reused).

$$F_{\text{net,cell}} = (F_{\text{cell,on}} - F_{\text{cell,off}}) \times g \times \frac{\ell_2}{\ell_1} \quad (12)$$

This method is noisier (turbulent plasma) but gives time-resolved thrust data.

7 Materials and Hardware

Component	Spec	Notes	Cost
Magnetron assembly	2.45 GHz, ≥ 800 W	Salvage from domestic microwave oven. Retain HV transformer, capacitor, fan, interlocks.	€0–€20
HV power supply (MOT)	≈ 2100 V, 300 mA	Retained from donor oven. Handle with extreme caution.	(from oven)
HV capacitor + diode	≈ 2100 V, $1\ \mu\text{F}$	Retain from oven; discharge before any handling.	(from oven)
Waveguide (handmade)	$90 \times 50 \times 600$ mm, Al	2 mm Al flat bar/sheet, bent and riveted. Flattened section $90 \times 25 \times 100$ mm. Short-circuit end plate.	$\approx$ €15
Three-stub tuner stubs	M8 SS316 bolts $\times 3$	Inserted through top wall of waveguide. Adjust for minimum reflected power.	$\approx$ €3
Quartz/borosilicate tube	$\varnothing_i = 24$, $\varnothing_o = 27$, L=600 mm	Borosilicate for initial tests; upgrade to quartz if needed.	$\approx$ €15
Vortex air inlet fitting	SS316 collar + 6 mm Cu elbow	Tangential injection; fabricate from bent copper tube + SS316 collar bored to tube OD.	$\approx$ €5
Air compressor	≥ 30 L/min, 1–3 bar	University lab asset (loan).	€0
Rotameter	0–2 m ³ /h	Calibrate before test.	$\approx$ €20
Steel ball force gauge	hollow steel ball 100–200 g, $D > 28$ mm	Drill 5 mm hole on top; use M3 steel bolts as calibrated beads.	$\approx$ €5
Water load / circulator	1 L water container or ferrite circulator	Absorbs reflected power; protects magnetron.	$\approx$ €5
Cooling fan	from donor oven	Cool magnetron during operation.	(from oven)
Microwave leakage detector	portable 2.45 GHz detector	Check for leaks before each test session. Mandatory.	$\approx$ €20
Safety interlock box	door switches from oven, wired in series	System cannot energise unless box is closed.	(from oven)
Load cell + HX711 + Arduino	from Phase 3A	Optional Method 2 measurement.	(Phase 3A)
Phase 5 total (excluding Phase 3A hardware)			$\approx$€108

Table 4: ICER Phase 5 hardware list.

8 Experimental Procedure

8.1 Pre-Test Checks

- P1. Microwave leakage check.** With the system assembled, use the leakage detector to scan all waveguide seams and the quartz tube aperture **before** energising. Zero-leakage from seams; accept $< 5\text{ mW/cm}^2$ at 5 cm from aperture (ICNIRP limit for 2.45 GHz occupational exposure: 10 mW/cm^2).
- P2. Interlock check.** Verify door interlock: system must not power when interlock is open.
- P3. Capacitor discharge.** Verify HV capacitor is discharged with a $10\text{ k}\Omega$ / 10 W bleed resistor connected across terminals before any mechanical intervention.
- P4. Air flow calibration.** Set rotameter to 0.7, 1.0, 1.3, $1.45\text{ m}^3/\text{h}$; record flow.
- P5. Steel ball baseline (F_0).** Set air flow; add beads until ball is stationary; record M_0 .

8.2 Test Procedure

- T1. Turn on cooling fan and water load.**

- T2. Set air flow** to first set point (e.g. $1.45 \text{ m}^3/\text{h}$).
- T3. Energise magnetron.** Close safety interlock; connect to mains. **Stand back minimum 0.5 m from the waveguide aperture.**
- T4. Ignite plasma.** The plasma ignites spontaneously when the microwave field is strong enough; if not, a brief spark from an igniter wire (from oven) at the tube opening initiates the arc.
- T5. Establish threshold weight M :** remove beads from steel ball (it will now vibrate) then add beads one by one until vibration stops. Record M .
- T6. Compute:** $F_{\text{net}} = (M - M_0)g$; $P_{\text{net}} = F_{\text{net}}/(\pi R^2)$.
- T7. Vary microwave power** by adjusting the variac on the MOT primary (if available) to 400, 600, 800 W. Repeat steps 5–6 at each power level.
- T8. Vary air flow** at fixed power (400 W). Repeat steps 5–6 at each flow: 0.7, 0.85, 1.0, 1.15, 1.30, $1.45 \text{ m}^3/\text{h}$.
- T9. Shut down.** De-energise magnetron; turn off air; allow 2 min cool-down. Discharge HV capacitor before any physical adjustment.

8.3 ABEP Simulation

To demonstrate the Skyraker Mode 3 (ABEP) principle, replace the compressor with a controlled leak from a small vacuum vessel, simulating low-density atmospheric gas ingestion:

- Partially throttle air flow to $<0.2 \text{ m}^3/\text{h}$ (very dilute regime);
- Record minimum power for sustained plasma at each flow rate;
- Plot $F_{\text{net}}/\dot{m}_{\text{air}}$ vs. $P_{\mu}/\dot{m}_{\text{air}}$: this characterises the ABEP figure of merit.

9 Data Table

Table 5: Phase 5 data acquisition sheet (after Ye et al. protocol).

P_{μ} [W]	$\dot{V}$ [m^3/h]	M_0 [g]	M [g]	F_{net} [N]	P_{net} [N/m^2]	F/P [N/kW]
0	0.70		–	–	–	–
0	1.45		–	–	–	–
400	0.70	(3A)				
400	1.00	(3A)				
400	1.30	(3A)				
400	1.45	(3A)				
600	1.15	(3A)				
800	1.00	(3A)				

M_0 (grey): measure once per air flow setting at $P_{\mu} = 0$; carry forward. $F_{\text{net}} = (M - M_0) \times 9.81 \times 10^{-3} \text{ N}$;
 $P_{\text{net}} = F_{\text{net}}/(\pi \times 0.012^2) \text{ N}/\text{m}^2$; $F/P = F_{\text{net}}/(P_{\mu} \times 10^{-3}) \text{ N}/\text{kW}$.

10 Safety

Microwave radiation: 2.45 GHz at 1 kW. Unshielded leakage causes deep-tissue heating without sensation – **no thermal warning**.

High voltage: MOT outputs ≈ 2100 V at 300 mA – lethal if contacted. HV capacitor retains charge after power-off.

Plasma at 1000 °C: contact or proximity to the open quartz tube is a severe burn hazard.

Standards: ICNIRP 1998 (EMF exposure); IEC 60335-2-25 (microwave appliances); CEI 64-8 (HV wiring); EN 60519-6 (microwave heating equipment).

- (a) **Microwave safety is the primary hazard.** Use a calibrated leakage detector before every test session. All waveguide seams must be covered with conductive tape and verified leakage-free.
- (b) **Safety interlock** (from donor oven) wired in series with MOT primary: system cannot energise when interlock is open. Non-negotiable.
- (c) **HV capacitor discharge** with a 10 k Ω /10 W resistor before any disassembly. Wait 30 s after discharge. **Never assume the capacitor is discharged.** Measure with a HV meter.
- (d) **Operator distance:** ≥ 1 m from the open end of the quartz tube during operation. A polycarbonate shield rated for plasma splash.
- (e) **Cooling:** magnetron must be fan-cooled during operation; never run > 30 s without airflow.
- (f) **Two-person rule.** Never operate HV/microwave equipment alone.
- (g) **Fire extinguisher** (CO₂, class C electrical) within 2 m.
- (h) **No flammable materials** within 0.5 m of the plasma tube.

11 Phase 5 Completion Criteria

Phase 5 pass: $F_{\text{net}} > 0$ measured at ≥ 3 power/flow combinations; F/P within 50% of Ye et al. reference (28 N/kW at 400 W, 1.45 m³/h); ABEP simulation: sustained plasma at $\dot{V} < 0.3$ m³/h demonstrated; no microwave leakage above ICNIRP limit measured during operation.

12 Budget Summary

Item	Source	Qty	Cost
Domestic microwave oven (salvage)	Second-hand market / university surplus	1	€0–€20
Aluminium flat bar 90×2 mm, 2 m	Ferramenta	1	≈€10
Aluminium sheet 50×2 mm, 2 m	Ferramenta	1	≈€5
Borosilicate tube $\varnothing_i = 24$, L=600 mm	Amazon / lab glass	1	≈€15
SS316 collar + 6 mm Cu tube (vortex inlet)	Hardware	1 set	≈€5
Rotameter 0–2 m ³ /h	Amazon	1	≈€20
Steel ball ($D \approx 76$ mm)	Hardware / online	1	≈€5
M3 steel bolts (calibrated beads)	Hardware	1 pack	≈€3
Microwave leakage detector	Amazon	1	≈€20
10 k Ω /10 W discharge resistor + HV probe	Amazon	1 set	≈€10
Conductive copper tape (waveguide sealing)	Amazon	1 roll	≈€8
Miscellaneous (rivets, screws, tape)	Hardware	–	≈€7
Total Phase 5			≈€108–€128
Phases 0–5 total (cumulative)			<€400

Table 6: Phase 5 budget. The dominant cost (magnetron) is nearly zero via salvage.

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